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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

First / Second Semester B.E. (Credit System) Degree Examinations

Supplementary Examinations – December 2025

21CV103 – ELEMENTS OF CIVIL ENGINEERING

20CV103 / 17CV103 – ENGINEERING MECHANICS / ELEMENTS OF CIVIL ENGINEERING AND ENGINEERING MECHANICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III.**

Unit – I

Marks BT* CO* PO*

1. a) Define equilibrium and mention the conditions of equilibrium.
 - i) coplanar concurrent force system
 - ii) coplanar non-concurrent force system
- b) Define free body diagram and explain with example.
- c) Determine the resultant of co-planar concurrent force system shown in fig. Q 1 (c). Compute the magnitude and direction with x-axis.

5 L*1 2 1,2
5 L2 2 1,2

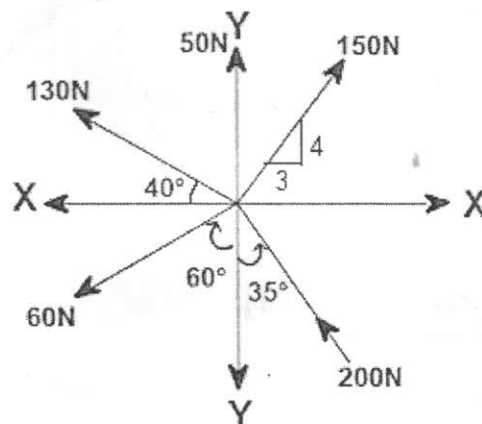


Fig. Q 1 (c)

10 L3 1 1,2

2. a) Explain the following scopes of Civil Engineering in the economic development of a nation, i) Transportation Engineering, ii) Water resources Engineering.
- b) Explain any five-force system with neat sketches.
- c) Sketch the resultant of force system acting on the plate element with respect to point 'O' shown in fig. Q 2 (c). Given the grid size 1m x 1 m.

5 L2 1 1,2
5 L2 1 1,2

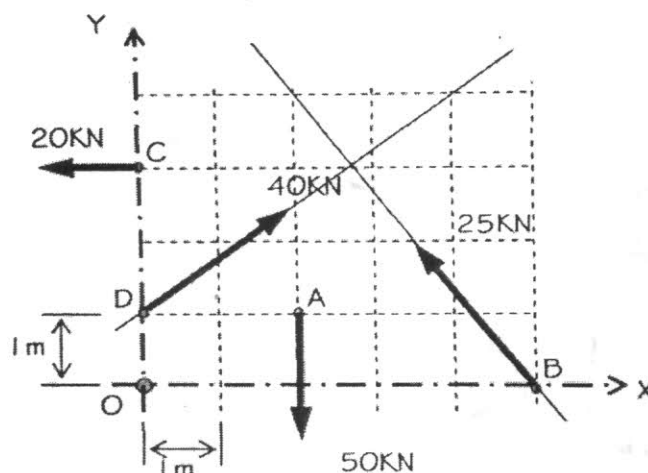


Fig. Q 2 (c)

10 L3 2 1,2

3. a) State and explain principle of transmissibility of a force with neat sketch. State its limitations
- b) List and explain the basic idealization in Engineering mechanics
- c) Two smooth spheres of weight 100N and radius 250 mm each are in equilibrium in a horizontal channel of width 870 mm shown in fig. Q 3 (c). Find the reactions at the surfaces of contact A, B, C, D assuming all surfaces to be smooth.

5	L2	1	1,2
5	L2	1	1,2

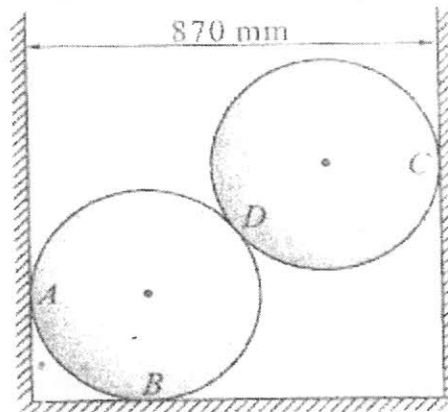


Fig. Q 3 (c)

10	L3	2	1,2
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Unit – II

4. a) Write any four Coulomb's laws of friction.
- b) Define i) limiting friction ii) coefficient of friction and ii) angle of repose.
- c) Determine the resultant reactions at the supports for the loaded beam shown in fig. Q 4 (c).

4	L1	4	1,2
6	L1	4	1,2

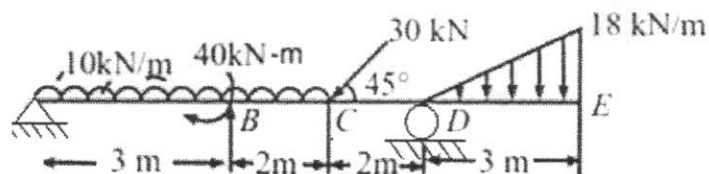


Fig. Q 4 (c)

10	L3	3	1,2
4	L2	3	1,2
6	L2	3	1,2

5. a) Distinguish uniformly distributed load and uniformly varying load.
- b) Explain any three types of supports with neat sketches.
- c) Find the least value of P required to cause the system of blocks impend shown in Fig. Q 5 (c). The coefficient of friction between the contact surfaces is 0.20.

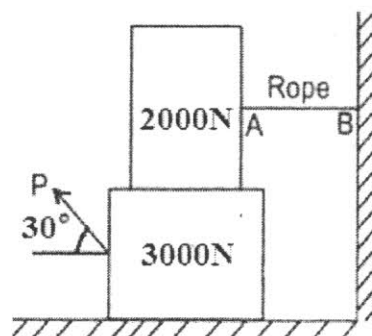


Fig. Q 5 (c)

10	L3	4	1,2
5	L3	4	1,2

6. a) Justify angle of repose is equal to angle of friction.
- b) A 400 N block is resting on a rough horizontal surface for which the coefficient of friction is 0.40. Determine the force P required to cause motion to impend if applied to the block downward at 30° with the horizontal.

5	L3	4	1,2
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- c) Determine the resultant reactions at the supports for the loaded beam shown in fig. Q 6 ©.

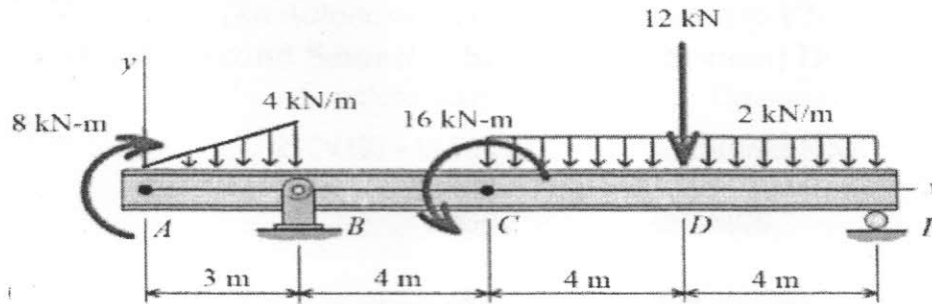


Fig. Q 6 ©

Unit – III

7. a) State perpendicular and parallel axes theorem with the expression.
b) Derive an expression for moment of inertia of a triangular lamina about its centroidal axes from the method of integration.
c) Locate the centroid of the shaded area about the axes shown in fig. Q 7 ©

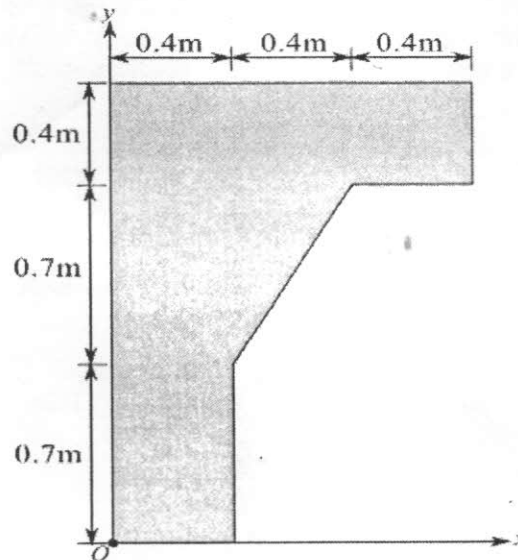


Fig. Q 7 ©

8. a) Distinguish centroid and center of gravity.
b) Determine an expression for centroid of quarter circular area about its base from the method of integration.
c) Determine the second moment of the shaded area of the section in fig. Q 8 (c) with respect to centroidal axes.

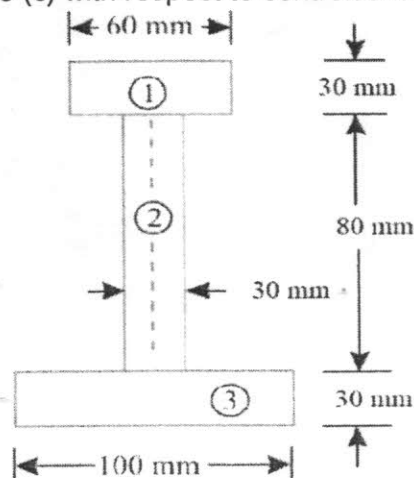


Fig. Q 8 (c)

10 L3 3 1,2

4 L2 5 1,2

6 L3 5 1,2

10 L3 5 1,2

4 L2 5 1,2

6 L3 5 1,2

10 L3 5 1,2